HACKATHON PROJECT

Can RSI improve an LLM's understanding of physics?

A stronger model improving a weaker one, tested on a ball and a wall.

BACKGROUND

Generated scenes now look strikingly real.

Precise objects from a prompt

GPT-6 Astra users report a detailed PlayStation 4 controller in about 12 minutes, and an Xbox Series X in 30.

X trending summary (Grok-generated), Sep 5, 2026

← Trending

GPT-6 Astra Generates Stunning SVG Controllers from Text Prompts

Last updated Sep 5, 2026

Released on September 3, 2026, GPT-6 Astra excels at computer-use tasks and creative generation, now rolling out to ChatGPT subscribers and API users. Users shared examples like a precise PlayStation 4 controller in 12 minutes, an Xbox Series X in 30 minutes, and a PlayStation 5 in 32 minutes, praising their realism and detail. Reactions mix awe at the vector-graphics leap with questions about editable code, highlighting the model's potential for assets like 3D models and game prototypes.

This story is a summary of posts on X and may evolve over time. Grok can make mistakes, verify its outputs.

EXAMPLE

`media/x_post_2.png`

another post, demo, or generated image

THE PROBLEM

But the motion often breaks the physics.

Newton's laws
Speed changes for no reason.

Energy conservation
A bounce returns more energy than went in.

Contact
Does the ball come back off the wall, or pass through?

Long videos
Errors accumulate, so physics can drift over time.

VIDEO

`media/phys_fail.mp4`

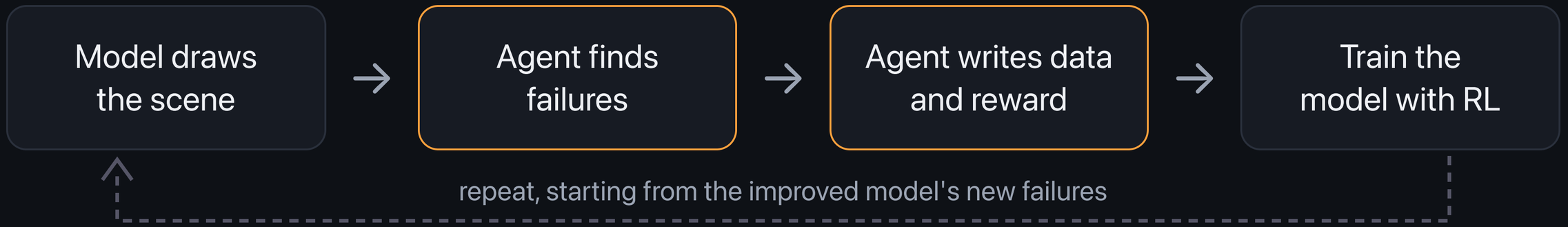
a generated video that breaks the physics

Our finding

Describe the video and what is wrong

OUR IDEA

Put an agent in the improvement loop.



- The agent, a stronger model (Claude), builds the training environment.
- It generates the data and tunes the reward.
- It fine-tunes the model automatically from the results, and explores on its own to raise the score.

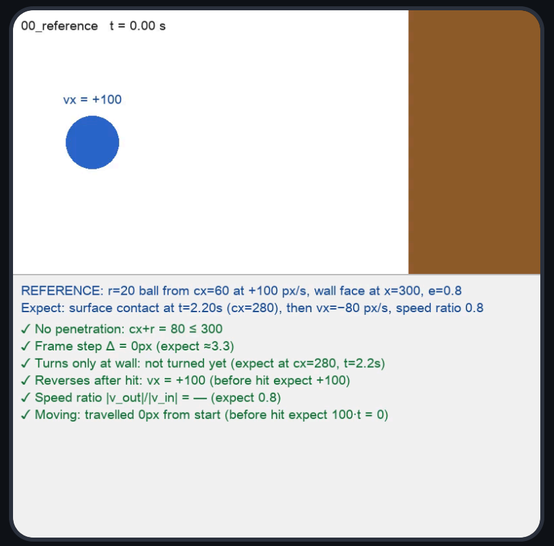
MODEL AND TRAINING

Small model, reinforcement learning.

MODEL	**Small Qwen instruct model** Qwen2.5-0.5B-Instruct
ALGORITHM	**GRPO with LoRA** Hugging Face TRL
REWARD	**Physics score from a checker** Written and revised by the agent each round
DATA	**Randomized ball and wall scenes** Generated by the agent

EXAMPLES

What correct physics looks like.



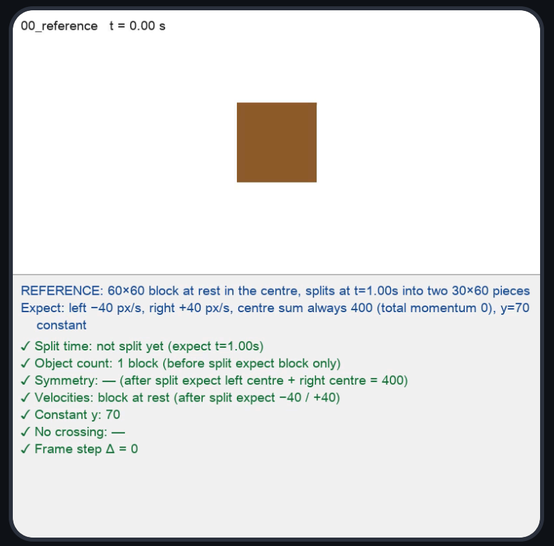
Ball hits a wall

Rebounds at 0.8× its speed



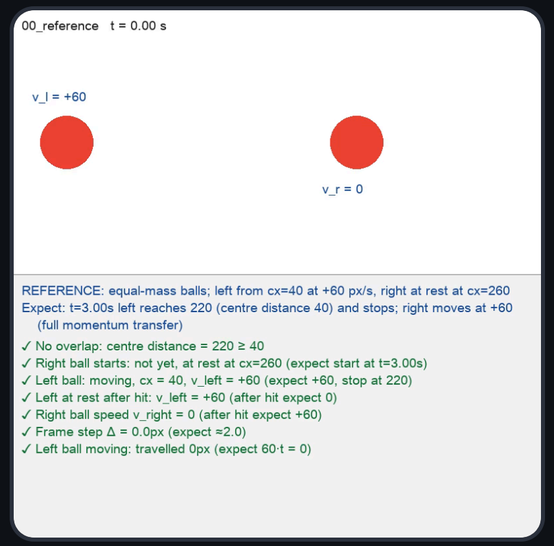
Ball bounces

Bounce heights decay



Block splits

Momentum stays zero

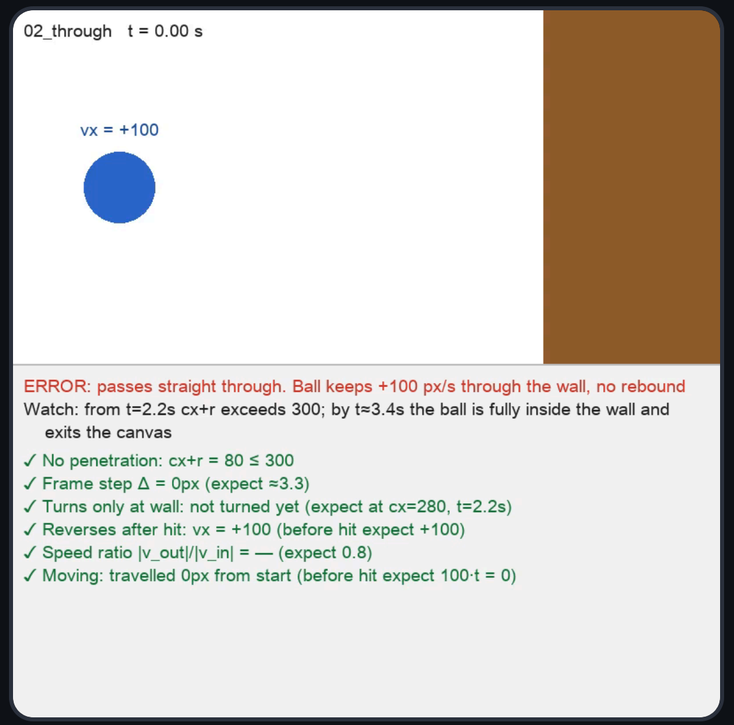


Two balls collide

Momentum passes on

FAILURE MODES

Each failure becomes a reward.

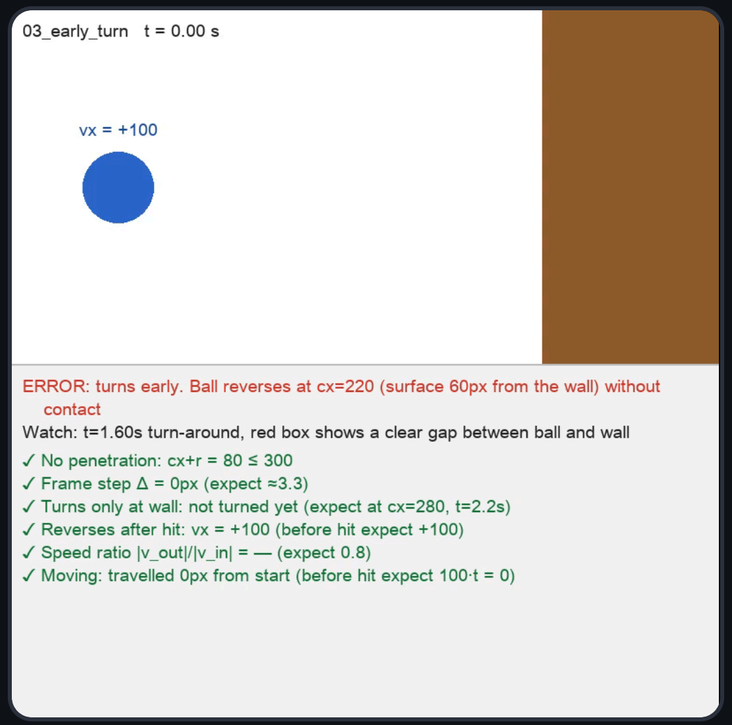


ERROR: passes straight through. Ball keeps +100 px/s through the wall, no rebound
Watch: from t=2.2s cx+r exceeds 300; by t≈3.4s the ball is fully inside the wall and exits the canvas

- ✓ No penetration: cx+r = 80 ≤ 300
- ✓ Frame step Δ = 0px (expect ≈3.3)
- ✓ Turns only at wall: not turned yet (expect at cx=280, t=2.2s)
- ✓ Reverses after hit: vx = +100 (before hit expect +100)
- ✓ Speed ratio |v_out|/|v_in| = — (expect 0.8)
- ✓ Moving: travelled 0px from start (before hit expect 100·t = 0)

Passes through the wall

→ penetration is a hard violation

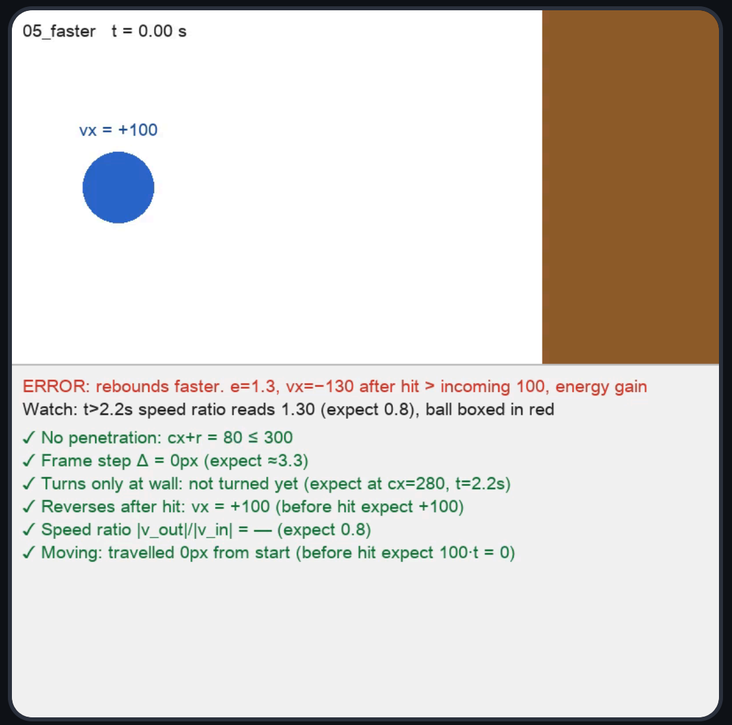


ERROR: turns early. Ball reverses at cx=220 (surface 60px from the wall) without contact
Watch: t=1.60s turn-around, red box shows a clear gap between ball and wall

- ✓ No penetration: cx+r = 80 ≤ 300
- ✓ Frame step Δ = 0px (expect ≈3.3)
- ✓ Turns only at wall: not turned yet (expect at cx=280, t=2.2s)
- ✓ Reverses after hit: vx = +100 (before hit expect +100)
- ✓ Speed ratio |v_out|/|v_in| = — (expect 0.8)
- ✓ Moving: travelled 0px from start (before hit expect 100·t = 0)

Turns back early

→ must reach the wall



ERROR: rebounds faster. e=1.3, vx=−130 after hit > incoming 100, energy gain
Watch: t>2.2s speed ratio reads 1.30 (expect 0.8), ball boxed in red

- ✓ No penetration: cx+r = 80 ≤ 300
- ✓ Frame step Δ = 0px (expect ≈3.3)
- ✓ Turns only at wall: not turned yet (expect at cx=280, t=2.2s)
- ✓ Reverses after hit: vx = +100 (before hit expect +100)
- ✓ Speed ratio |v_out|/|v_in| = — (expect 0.8)
- ✓ Moving: travelled 0px from start (before hit expect 100·t = 0)

Bounces back faster

→ energy gain is penalized

RESULTS

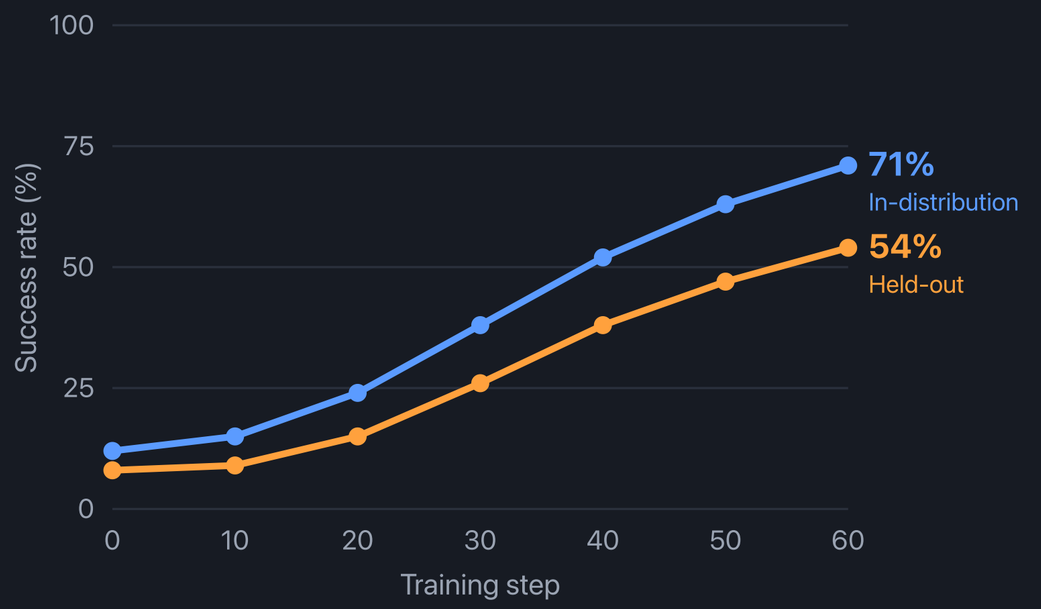
FAKE DATA · PREVIEW ONLY

Success rate on cases it never saw.

Held-out scenes, before → after training

8% → 54%

	IN-DISTRIBUTION	HELD-OUT
Base model	12%	8%
After training	71%	54%



Success: no physics rule broken, and contact time, bounce speed, and trajectory all accurate. 200 scenes per set.